

Johnny Dunn

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Summary

Full-stack engineer specializing in production AI/ML: RAG, LLM fine-tuning, classical deep learning, quantization, edge inference, and evaluation harnesses. Builds low-latency conversational voice interfaces with streaming STT/TTS, VAD, diarization, and turn-taking.

Professional Experience

Machine Learning Engineer

Independent Consulting - Contract | Remote | May 2019 – March 2026

- Built real-time edge AI surveillance on embedded SBC based on RK3588 (3 NPUs) in Go for Aye Eye Global (3+ months). Multi-cam RTSP, YOLOv8n detection, facial recognition, thermal throttling, with React UI, shipping for 5k+ units in 2026.
- Built unified ML pipeline in Python for Linkby (6-month project) automating feature engineering and model training from SQL for advertising data, enabling parallel training of RF, SVM, Naive Bayes, and LR models with A/B testing framework.
- Improved AI text detection accuracy from 55% to 71% for Grapple Research (5-month contract) by fine-tuning GPT-3.5 on expanded, deduplicated dataset. Reduced training costs by 64% with truncation & summarizing based on article structure.
- Developed document matching tool for ImmenseX (4-month engagement) reducing initial prompt writing by 60% for devs. Deployed RAG pipeline using Weaviate, Neo4j (hybrid vector and graph-based search) for conversational memories.
- Achieved 77% sentiment analysis accuracy (random forest) trained on 10K texts for HereAfterLegacy.ai; built out initial MVP to scrape scripts from TV shows to recreate digital persona of 3 fictional characters (14-month contract).
- Trained neural style transfer models for public exhibit applying film-inspired filters to selfies for Edelman (3-month project).
- Created user research, animated wireframes, and functional demo for MVP at Notebrush (15-month engagement).

Senior Blockchain Engineer

Tilting Point | New York, NY | Nov 2021 – Feb 2023

- Developed ERC-20/721 smart contracts with upgradability / proxy patterns, built and deployed backend REST APIs for NFT marketplace, and internal airdropping tool. Rescued \$1M MATIC through proxy contract methods regenerating Gnosis safes.
- Architected blockchain state gRPC microservices, supporting 4+ games with 1000+ events/sec (on/off-chain throughput).

Software Engineer 2

eBay Inc. | New York, NY | Jun 2020 – Apr 2021

- Built REST APIs in Scala / Spring-Boot showing new ads in real-time from clicking selected item options (in under 1s latency).
- Handled E2E ML delivery infrastructure (K8s + Jenkins) enabling teams to deploy models into sandboxed UIs; accelerated deep learning team iteration by 50% allowing ads team to validate changes without waiting for releases from other teams.

Cognitive Implementation Engineer

Amelia.ai | New York, NY | Jun 2017 – Dec 2017

- Implemented features like PII redaction for HIPAA, SOC 2 compliance for AI assistant serving Fortune 500 medical suppliers.

Projects

AgentOS – TS agentic orchestration; guardrails presets (PII redaction) & graph-based, multimodal RAG. github.com/framersai/agentos

Quarry – Fully offline notetaking app that organizes itself with NLP and can answer your questions. github.com/framersai/quarry

tenets – MCP server for coding agents to aggregate context for prompts with neural ranking, fully offline. github.com/jddunn/tenets

Restless - Hierarchical attention network model for anti-malware; 93% accuracy, [ClAMP](https://arxiv.org/abs/2305.12202) dataset. github.com/jddunn/restless

Education

Bachelor of Fine Arts in Game Design (May 2017) – Parsons School of Design

Awards

5th Place, AT&T National VR and AR Challenge, with THiNKIMMERSIVE team. (<http://npe100.com/> / mobile app), 2017.

Skills

Programming Languages	Python, TypeScript, JavaScript, Go, Scala, SQL, Solidity
Frontend	React, React Native, Next.js, Redux, Vue, Nuxt, Tailwind, D3.js, three.js
Backend	FastAPI, Django, Node.js (Express, NestJS), gRPC, GraphQL, RabbitMQ, Kafka, Stripe
AI / ML	PyTorch, TensorFlow, LangGraph, LlamaIndex, RAGAS, DeepEval, Hugging Face, Transformers, sklearn, spaCy, Pandas, NumPy, OpenCV, CUDA, XGBoost, Stable Diffusion, DSPy, promptfoo
Databases / DevOps	PostgreSQL, SQLAlchemy, Supabase, Redis, Neo4j, Pinecone, Weaviate, MongoDB, Elasticsearch, AWS (Lambda, EC2, EBS, Bedrock), Docker, Kubernetes, Jenkins, GitHub Actions
Tools / Platforms / Other	Unity3D, Adobe Creative Suite, Electron, Capacitor, Android Studio, Quasar, Selenium, Puppeteer